

Branchseed Challenge: Winning Technical and Product Strategy

Executive decision

Build a mask-anchored ostium detector with short, constrained vessel tracking, not a full vascular segmentation model. The core should crop tightly around the supplied aorta, learn each scan's contrast statistics from the aortic lumen, score outward continuations across the aortic wall, track only the first 10 mm of each candidate, and use graph-based deduplication to return one instance per true direct daughter. Pair it with an offline Aorta Map: an unwrapped view of the aortic surface linked to CT slices, a translucent 3D model, and one evidence card per branch.

This choice follows the score: branch discovery is 45%, ostium localization is 25%, daughter quality is 15%, and efficiency plus reproducibility are 15%. It also follows the actual task. The official document specifies contrast-enhanced CT/CTA, not MRI; it explicitly excludes anatomical naming, parent-aorta segmentation, complete distal-tree reconstruction, and invisible-branch prediction.^[1]

The winning system should therefore spend almost all engineering effort on five things: recall without duplicates, millimetre-correct geometry, cropped CPU execution, auditable confidence, and a five-minute demo that makes every detection visually defensible.

1. Hackathon requirements

1.1 Exact task

Given a CT volume and a binary mask containing only the parent abdominal-aortic lumen, automatically discover every eligible artery whose lumen connects directly to that parent. Each real daughter must be returned exactly once as a separate instance. For every instance, return the ostium centre, a daughter-lumen centre point 5 mm outward along the vessel path, the local radius at that seed, and a unit direction vector pointing from the ostium into the daughter.^[1]

The task is instance discovery and proximal geometry, not general vessel segmentation. The branch can be followed for at most 10 mm beyond the ostium or until its first downstream bifurcation, whichever comes first. A branch is eligible only if its contrast-filled lumen remains followable for at least 5 mm and its origin meets a minimum size that the organisers have not yet supplied.^[1]

1.2 Explicit requirements versus proposals

Status	Requirement or idea	Consequence
Explicit	Input is CT/CTA plus parent-aorta binary NIfTI mask	Correct the project language from MRI to CT immediately
Explicit	Image and mask share a grid and physical coordinate system	Validate both before inference; fail clearly on mismatch
Explicit	Detect a variable number of direct daughters	Never assume a fixed anatomy list or branch count
Explicit	Return one instance per real daughter	Duplicate suppression is a first-class algorithmic stage
Explicit	Report physical millimetres using SimpleITK coordinates	Never emit voxel indices or hand-written affine math
Explicit	Seed is 5 mm along the daughter path	Measure arc length, not straight-line displacement
Explicit	Direction is a unit vector into the daughter	Fit a stable tangent and assert norm = 1
Explicit	Radius is local at the daughter seed	Measure in a plane normal to the proximal centreline
Explicit	Exclude cropped superior/inferior aorta ends	Detect and suppress cap surfaces
Explicit	Common trunk = one origin; separate wall openings = two	Cluster by ostium connectivity, not downstream branch count
Explicit	Iliac terminal division is optional, separate from core	Keep it behind an extension flag
Explicit	CPU-only: 4 cores, 8 GB RAM, no internet	Crop early; use float32; bundle every dependency and weight
Explicit	Initial target <=60 seconds average per case	Benchmark end-to-end, including I/O unless organisers say otherwise
Explicit	Required CLI signature	Preserve it exactly in <code>run.py</code>
Explicit	Visual check for at least three cases	Produce deterministic verification renders from inference outputs
Explicit	Five-minute demo including runtime and failures	Show evidence and limitations, not just a polished mesh
Proposed	Confidence score and evidence sidecar	Keep outside the strict submission JSON unless extras are confirmed
Proposed	Unwrapped Aorta Map and linked branch cards	Best product differentiator; it does not alter scored output
Proposed	Lightweight candidate reranker	Add only after a strong classical baseline and reference labels exist

1.3 Required output and safest contract

Use the exact JSON fields shown by the organisers: `case_id`, `parent_instance_id`, and a `daughters` array containing `instance_id`, `parent_instance_id`, `ostium_xyz_mm`, `seed_xyz_mm`, `radius_mm`, and `direction_xyz`. Sort detections deterministically by aortic centreline position and circumferential angle, then assign `branch_001`, `branch_002`, and so on. If no branch survives eligibility checks, emit an empty list.^[1]

Although the PDF says another creative format is allowed, the supplied run command ends in `prediction.json`. The safest interpretation is: always write strict JSON for scoring; write creativity to separate diagnostic files. Do not risk an evaluator rejecting extra fields.

1.4 Scoring priorities

Category	Weight	Engineering implication
Branch discovery	45%	Optimize candidate recall first, then use path evidence and deduplication to protect precision
Ostium localization	25%	Refine the path/surface intersection at sub-voxel precision in physical space
Daughter-instance quality	15%	Compute the 5 mm seed, tangent, and cross-sectional radius from one coherent short path
Compute efficiency	10%	Operate in a narrow periaortic ROI and avoid full-volume neural inference
Reproducibility	5%	Exact CLI, deterministic results, pinned offline environment, schema and geometry tests

The evaluation description does not provide the distance tolerance for a correct match, the localization scoring curve, the direction angular tolerance, the radius error function, or the exact one-to-one assignment method. These are material unknowns because they determine how aggressively to suppress close candidates.^[1]

1.5 Dataset audit

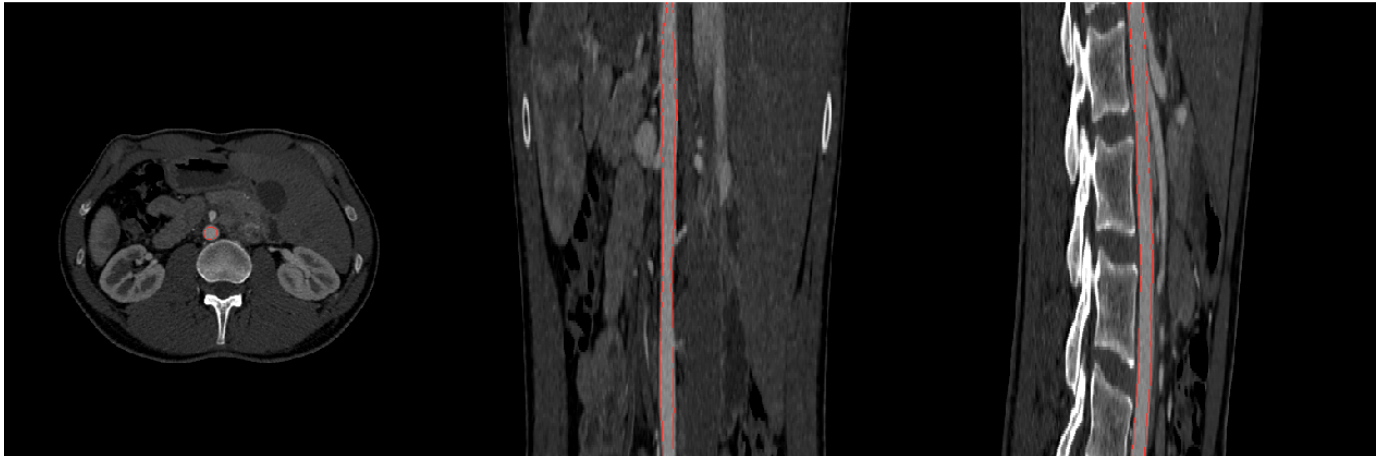
The supplied ZIP contains 25 paired cases and no daughter-reference output files. Every case has one `orig*.nii` CT and one binary `mask*.nii` aorta mask. All image/mask pairs have matching shapes, spacings, and s-form transforms in the files inspected.^[2]

Observed property	Result	Design consequence
Cases	25 paired volumes	Too small for training a robust 3D model from scratch, especially without daughter labels
Shapes	25 distinct shapes	No fixed tensor size assumptions; crop dynamically
Voxel counts	4.61M to 54.26M	Full-volume multi-scale filtering risks memory and time overruns
Spacing	0.625-1.5 mm in-plane; 0.8-1.5 mm through-plane	All radii, path lengths, dilation, and suppression distances must be spacing-aware
Aortic median CT value	Approximately 80-564 HU across cases	A fixed global HU threshold will be brittle
Aorta-mask volume	Approximately 7.1-197.5 mL	Coverage varies substantially; cap handling is essential
Mask values	Binary 0/1	No daughter labels are embedded in the supplied masks
Storage cohorts	Subjects 1-15 are uncompressed <code>.nii</code> ; 16-25 are gzip streams named <code>.nii</code>	Add magic-byte file sniffing and a regression test for both encodings
Geometry cohorts	Subjects 1-15 mostly submillimetre x/y and 0.8 mm z; 16-25 are 1.5 mm isotropic	Tune on both cohorts; expect partial-volume degradation in the latter
Header cohorts	Subjects 1-15 use q-form and s-form code 1; 16-25 use s-form code 2 and q-form 0	Trust the image library's physical transform; never assume an orientation

subject001 axial

subject001 coronal

subject001 sagittal



subject024 axial

subject024 coronal

subject024 sagittal



Representative central views from a high-resolution/high-enhancement case (subject001) and a coarse/low-enhancement case (subject024). Red outlines show the supplied parent-aorta mask; the contrast and resolution shift is visible even before algorithmic processing.

The mixed resolution and contrast distributions are the strongest evidence for per-case, mask-derived normalization. Use an eroded aortic core to estimate enhanced-blood statistics, and use a periaortic shell to estimate competing tissue/background statistics. Preserve both source geometry and a direct round-trip test using SimpleITK's image transform methods.^[3]

1.6 Ambiguities to resolve with organisers

Rank these questions by urgency:

1. What is the minimum eligible ostium size or daughter radius?
2. Where are the promised development reference outputs, and which cases are development versus hidden evaluation?
3. What exact tolerances and formulas are used for matching, localization, direction, seed-on-vessel, and radius?
4. Is the 60-second target inclusive of loading, decompression, visualization, and JSON writing?
5. Will the final input use .nii, .nii.gz, or both, and is the gzip-content/.nii-suffix mismatch intentional?
6. May the submission use external public data and pretrained weights? Must licenses be included?
7. Are additional JSON fields accepted, or must diagnostics be separate?
8. Should ostia and seeds be emitted at voxel centres or may continuous physical coordinates be used?
9. Is iliac detection excluded from the core score by a flag or by reference construction?
10. What CPU model and operating system will be used, and may native compiled wheels be bundled?

Until answered, make every uncertain threshold configurable, keep the core JSON minimal, and optimize against two internal operating points: a recall-biased development mode and a precision-balanced submission mode.

2. Technical approach landscape

2.1 Comparative decision table

Scores are judgments for this challenge, not published performance claims. Five is best; risk is reversed, so five means high risk.

Approach	Accuracy potential	CPU fit	Label need	Prototype time	Risk	Decision
Fixed HU threshold + connected components	2	5	None	Hours	4	Baseline only
Adaptive region growing from aortic wall	3	5	None	<1 day	3	Speed fallback
Multi-scale Hessian vesselness in cropped ROI	3	4	None	1 day	3	Useful feature, never sole detector
Surface-normal projection + local maxima	4	5	None	1-2 days	2	Primary candidate generator
Short beam search / fast marching tracker	4	4	None	1-2 days	2	Primary candidate validator
Segmentation + skeleton + graph pruning	4	3	Masks needed or pseudo-labels	2-4 days	4	Secondary pipeline only
Small 2D/2.5D U-Net in periaortic patches	4	4	Daughter labels	2-4 days	4	Optional if references arrive
Full 3D nnU-Net from scratch	5 in a large labelled setting	1	Dense labels	Many days	5	Reject for core
TotalSegmentator / MONAI whole-body model	2 for this exact output	1-2	Pretrained	Hours to test	5	Dev-time benchmark only
MedSAM / SAM-style prompting	2	2	Prompts or automatic prompt model	1-3 days	5	Reject for autonomous core
Tiny learned candidate reranker + classical tracker	5 if labelled candidates exist	5	Sparse candidate labels	1-2 days	3	Best accuracy upgrade

2.2 Classical thresholding and region growing

The simplest pipeline dilates the aorta, thresholds bright voxels in the shell, keeps connected components touching the aortic boundary, and fits a short centreline. It is fast and makes a valuable baseline, but it will leak into adjacent enhanced structures, fracture low-contrast branches, merge close ostia, and respond to calcification or bone. The observed 80-564 HU aortic median range rules out one global threshold.^[2]

Improve it by modeling intensity relative to the current scan: a robust aortic-core median and MAD, a local periaortic background model, hysteresis thresholds, spacing-aware morphology, and a minimum 5 mm path criterion. This remains the best speed-first fallback because every step is explainable and CPU-friendly.

2.3 Vesselness filters

Hessian-based multi-scale vesselness enhances bright tubular structures using eigenvalues of the image Hessian.^[4] It is appropriate as one feature in the periaortic shell. It is not sufficient by itself: junctions can produce gaps or unstable responses, and bright non-vessel structures can survive. Published CTA centreline work explicitly reports junction gaps and failures when intensity varies because of artifacts or calcification.^[5]

Use two or three physical scales matched to plausible daughter radii, computed only inside a cropped ROI. Fuse vesselness with intensity continuity, surface-normal alignment, and cross-sectional symmetry. If vesselness consumes more than roughly one quarter of the runtime budget or does not improve held-out branch F1, remove it from the submission and keep it as a visualization/debug layer.

2.4 Aortic-surface projection

This approach is unusually well matched to Branchseed. Estimate the aortic wall from the supplied mask, compute outward normals from a signed distance field, and sample short cylinders or cones extending outward. Score each surface location by sustained contrast-lumen likelihood, tubular symmetry, and vesselness. A published CTA aortic-root method detected coronary ostia by projecting average intensity along short cylinders from the segmented aortic surface and searching the resulting surface map for local maxima.^[6]

The important adaptation is to avoid any fixed number or anatomical prior. Generate all plausible maxima, exclude cap surfaces, and let short-path validation determine eligibility. This method is fast because it converts a full 3D search into a set of local outward tests around a known parent.

2.5 Short constrained tracking

The output itself asks for a short centreline-like object: origin, point at 5 mm, direction, and radius. Therefore track only 10 mm. Starting just outside each candidate ostium, use a physical-space beam search or local fast marching over a cost that rewards aorta-like contrast, vesselness, radial boundary evidence, and direction continuity while penalizing curvature, background, and travel back into the aorta. ITK's fast marching implementation solves an Eikonal propagation problem and supports early stopping, which is useful for small ROIs.^[7]

A beam width of three to five hypotheses is often a better hackathon tradeoff than a global shortest path because the endpoint is unknown. The local tracker literature shows that direction and radius can be predicted or scored from patches while tracing a vessel; published coronary results are encouraging but are not directly transferable to abdominal branches or this dataset.^[8]

2.6 Segmentation, skeletonization, and graph reconstruction

If a reliable binary periaortic vessel segmentation exists, skeletonization and graph decomposition make direct-daughter logic natural: subtract the supplied aorta, identify graph edges touching its surface, prune spurs, and stop at the first bifurcation. Topology-aware metrics such as cIDice were designed because overlap alone does not preserve vascular connectedness.^[9]

The difficulty is obtaining a clean segmentation. Threshold leaks create false graph edges; gaps delete real branches; skeletonization near the wide aortic junction is unstable. Use graph processing after local evidence generation, not as a promise that a weak segmentation will become correct. VMTK is excellent for centreline and radius analysis once a valid tubular surface exists, but producing that surface automatically is the hard part here.^[10]

2.7 Deep learning from scratch

Modern aortic segmentation competitions show that supervised 3D models can be powerful. AortaSeg24 provides 100 CTA volumes with 23 branch/zone labels, and leading solutions commonly used cascaded nnU-Net variants, targeted losses, and extensive augmentation.^[11] One official nnU-Net submission trained a large residual-encoder model on an A100 40 GB GPU.^[12]

That evidence does not make a full 3D network the right Branchseed core. The provided 25 cases do not include daughter masks, the target is variable unnamed origin instances rather than a fixed semantic label set, the evaluation is 4-core CPU with 8 GB RAM, and time is limited. Training from scratch is therefore a high-risk detour. If external training is allowed, public AVT data supplies 56 CTA vessel-tree masks, but its full-tree binary labels still need conversion into this task's direct-origin targets and may not match the hidden cohort.^[13]

2.8 Pretrained models and foundation models

TotalSegmentator. It supports CT on CPU and recommends `--fast` or ROI subsets for CPU execution.^[14] Its licensed `renal_arteries` task predicts celiac trunk, SMA, and renal arteries, not every unnamed eligible direct daughter; it also imposes a model-weight and license/offline-deployment burden.^[15] Use it only as a development comparison or pseudo-label source if rules and licensing permit.

MONAI. MONAI is a strong engineering framework, not a direct answer. Its whole-body CT bundle includes an aorta class but outputs 105 channels and documents GPU-heavy training/evaluation; it does not produce generic direct-daughter instances.^[16] Use MONAI only if the team already knows it and decides to train a cropped model.

nnU-Net. It is the strongest default for a genuinely labelled segmentation problem and supports CPU inference, but official guidance still recommends GPU for speed and describes CPU training as impractical.^[17] It becomes attractive only if dense daughter masks or external compatible data are available and a small cropped configuration can meet the CPU budget.

MedSAM/SAM-style systems. The official MedSAM inference flow relies on bounding-box prompts, while this challenge prohibits manual point placement and requires a variable number of automatically discovered branches.^[18] Adding an automatic prompt generator recreates the hard detection problem. Reject it for the scored core.

2.9 2D, 2.5D, and 3D comparison

Design	Strength	Weakness	Best use here
2D axial	Fast, simple, many samples	Loses oblique continuity; z-spacing varies	Candidate heatmap only
2.5D adjacent slices	Good CPU compromise; some continuity	Still weak for branches nearly parallel to slice plane	Learned candidate reranker
Small 3D patch model	Captures direction, lumen shape, and junction	Needs labels; slower but bounded by crop	Accuracy upgrade after classical proposals
Full-volume 3D	Highest context	Highest memory/time; unnecessary context	Reject

The most defensible learned option is a tiny 3D or 2.5D candidate classifier/regressor, not a segmentation network. Train it on hard negatives from the aortic surface and positives around reference ostia, then keep the deterministic tracker and geometry code. Quantize or export to ONNX only after measuring accuracy and latency.

3. Best candidate pipelines

3.1 Pipeline A - Surface proposals + local tracker + graph NMS

How it works. Crop to a 20-25 mm physical margin around the aorta. Build a signed distance field and surface normals. Calibrate lumen likelihood from the eroded aorta. Score short outward cones at surface points. Cluster maxima into candidate openings. Track each candidate 10 mm using beam search over a fused cost. Reject paths shorter than 5 mm or with poor cross-sectional evidence. Refine the ostium at the signed-distance zero crossing, compute the 5 mm seed and tangent, estimate radius, then deduplicate through an ostium/path-overlap graph.

Expected strengths. Excellent task alignment; no training labels; bounded CPU cost; direct control of duplicates; interpretable evidence; natural handling of variable branch count.

Main failures. Very low contrast, metal artifacts, aortic-mask boundary errors, tiny branches near the unknown size cutoff, and adjacent bright structures that mimic a short tube.

Hackathon judgment. Best overall pipeline and best risk-adjusted probability of completion.

3.2 Pipeline B - Pipeline A + tiny learned reranker

How it works. Keep Pipeline A's proposal generator. For each candidate, create a canonical patch aligned to the local aortic normal and tangent. A small 2.5D CNN or gradient-boosted classifier scores true ostium versus hard negative and optionally regresses a direction/radius correction. The deterministic tracker still generates geometry and enforces eligibility.

Data need. Reference ostia or carefully reviewed pseudo-labels. Candidate-level labels are much cheaper than dense vessel masks. Augment rotations around the normal, intensity/contrast, noise, blur, and 1.5 mm partial-volume conditions.

Expected strengths. Better precision at bone, calcification, venous enhancement, and irregular mask boundaries while preserving a fast candidate-limited workload.

Main failures. Overfitting to few annotated cases and domain shift between the two observed resolution cohorts.

Hackathon judgment. Best accuracy-first approach if references arrive early; otherwise do not attempt it.

3.3 Pipeline C - Adaptive shell segmentation + connected components

How it works. Dilate the aorta 12-15 mm, subtract the parent mask and cap zones, compute a case-adaptive lumen threshold, apply hysteresis and morphology, keep components connected to plausible wall openings, fit a PCA line or skeleton to the first 10 mm, and deduplicate components.

Expected strengths. Very fast, easy to debug, low dependency burden, and likely to satisfy the CLI and geometry contract quickly.

Main failures. Leaks, merged origins, discontinuities, and unstable radii at coarse resolution.

Hackathon judgment. Best speed-first fallback and essential first baseline.

3.4 Weighted comparison

Pipeline	Discovery 45	Ostium 25	Quality 15	Efficiency 10	Repro 5	Weighted score / 5	Delivery risk
A: surface + tracker + graph	4.3	4.4	4.2	4.2	4.8	4.34	Low-medium
B: A + learned reranker	4.7	4.5	4.4	3.8	4.0	4.48	Medium-high without labels
C: adaptive shell baseline	3.3	3.2	3.0	4.9	4.8	3.48	Low
Full 3D pretrained/finetuned	3.6	3.2	3.7	1.4	2.8	3.17	High

These scores are planning judgments. They are intended to expose tradeoffs, not claim measured performance.

4. Visualization ideas

The official requirement is only a visual check for three cases, but a carefully designed interface can differentiate the project without endangering the core. The interface should prove each machine-readable answer and support rapid clinical verification.

Rank	Visualization	Surgeon value	Judge impact	Difficulty	Recommendation
1	Unwrapped Aorta Map: centreline distance vs circumferential angle with ostia dots	5	5	3	Winning differentiator
2	Linked branch evidence card: axial/coronal/sagittal CT, ostium, 5 mm seed, arrow, radius ring	5	4	2	Must build
3	Translucent 3D aorta with coloured proximal stubs and direction arrows	4	5	2	Must build

Rank	Visualization	Surgeon value	Judge impact	Difficulty	Recommendation
4	Clickable branch table with coordinates, radius, path length, and confidence	5	3	1	Must build
5	Orthogonal cross-section at the 5 mm seed with fitted lumen boundary	5	4	2	High-value addition
6	Confidence/evidence view: intensity, vesselness, continuity, duplicate margin	4	4	2	High-value addition
7	Threshold-stability halo showing how much the origin moves under perturbation	4	4	3	Strong if time permits
8	Animated traversal of each 0-10 mm path	3	4	2	Demo polish
9	Diameter profile over only the proximal 10 mm	3	3	2	Useful but secondary
10	Full distal vascular tree	2	5	5	Reject; out of scope and risky
11	Automated stenosis/aneurysm diagnosis	2	4	5	Reject; unsupported clinical claim
12	CFD or virtual stent simulation	2	5	5	Reject; time sink unrelated to score

4.1 The Aorta Map

Create a straightened/unwrapped representation of the aortic surface. The vertical coordinate is arc length along the supplied aorta; the horizontal coordinate is angle around a parallel-transport frame. Plot each detected ostium as a numbered circle sized by estimated daughter radius and coloured by confidence. Clicking it selects the corresponding branch evidence card and 3D stub.

This is more useful than a generic rotatable model because it converts a difficult 3D relationship into a compact map of where openings are along and around the parent vessel. Published aortic-root work already demonstrates the utility of mapping surface-derived intensity in centreline/angle coordinates for ostium localization.^[6] The product innovation is linking that representation to traceable machine evidence rather than asserting anatomical names.

4.2 Branch evidence cards

Each branch card should answer six questions at a glance: Where is the origin? Does a bright lumen continue for 5 mm? Which way does it go? Where was radius measured? How confident is the detector? Is this a direct daughter rather than a downstream branch? Use synchronized slice views, a short coloured path, an origin crosshair, a 5 mm ring, and a small score breakdown.

This is both a clinician-facing verification workflow and a judge-facing explanation of the algorithm. It turns failure disclosure into a strength: low-confidence branches are visibly reviewable.

4.3 Visualization stack

Generate the scored JSON first. From the same result, emit an offline HTML report with Plotly 3D meshes/lines plus pre-rendered CT panels. Plotly can create interactive 3D meshes and can bundle its JavaScript for offline use.^[19] Keep the HTML generator outside timed inference unless the organisers explicitly include it.

For development and ground-truth review, use 3D Slicer with SlicerVMTK. Its modules support centreline extraction, branch decomposition, maximum-inscribed-sphere radius, orthogonal cross-sections, area, diameter, curvature, torsion, and tortuosity.^[20] Do not make Slicer a runtime dependency of the submission.

5. Winning differentiators

5.1 Ranked differentiators

1. Evidence-first Aorta Map. High wow factor, directly tied to the scored output, and clinically interpretable.
2. Topology-aware instance logic. Explicitly model wall openings, common trunks, nearby separate ostia, cap exclusions, and downstream bifurcations. This is more sophisticated than a segmentation screenshot and attacks the 45% discovery score.

3. Self-calibrating contrast model. Show that thresholds adapt to each case using the supplied parent mask. This is technically defensible against the measured 80-564 HU aortic median range.
4. Auditable confidence. Combine proposal strength, path persistence, radius consistency, and separation from the next-best hypothesis. Display confidence but do not allow it to replace hard eligibility logic.
5. Geometry contract validator. Automatically prove coordinate round-trips, 5 mm arc length, unit direction, positive radius, uniqueness, cap exclusion, and deterministic ordering. This quietly protects 45% of the score.
6. Two-resolution robustness. Demonstrate high-resolution and 1.5 mm cases side-by-side, including low-contrast examples. Judges will see deliberate domain-shift handling rather than cherry-picked beauty shots.
7. Optional learned reranker. Valuable only after it wins an ablation against the deterministic baseline.

5.2 Weak ideas and time sinks

- Anatomical vessel naming: explicitly out of scope and introduces avoidable false certainty.
- Full 3D model as the main result: visually familiar and not evidence of accurate origins.
- Complete distal-tree reconstruction: unnecessary, difficult, and penalizes runtime.
- Pathology claims: the challenge does not provide disease labels or ask for diagnosis.
- Population-normal comparisons: no demographic or validated normative dataset is supplied.
- CFD, stent placement, or intervention simulation: high engineering cost and weak score connection.
- LLM-generated clinical narratives: distract from geometry and create safety/credibility risk.
- Large model ensembles: incompatible with the CPU budget and reproducibility target.

6. Recommended architecture

NIfTI CT + aorta mask -> validation and crop -> adaptive lumen model -> aortic surface proposals -> short multi-hypothesis tracking -> ostium graph and deduplication -> 5 mm geometry and radius -> strict JSON -> offline Aorta Map and branch cards

6.1 Stage-by-stage design

Stage	Recommended method/tool	Input -> output	Effort	Main risk	Backup
1. I/O and geometry	SimpleITK; magic-byte sniffing wrapper	NIFTI files -> images with validated physical metadata	0.5 day	Mislabeled gzip content	Normalize suffix to temporary file, then read
2. Mask QA and crop	Largest component, spacing-aware morphology, 20-25 mm physical bounding box	Aorta mask -> clean mask + compact ROI	0.5 day	Mask defects or clipped coverage	Preserve original mask and log QC warnings
3. Per-case normalization	Eroded-core robust median/MAD plus periaortic background shell	CT + mask -> lumen-likelihood volume	0.5 day	Low enhancement or calcification	Two-component robust mixture + gradient evidence
4. Aortic geometry	Signed Maurer distance, gradient normals, lightweight centreline/frame	Mask -> surface points, normals, cap zones	1 day	Noisy normals at coarse spacing	Smooth distance field or use local ellipsoid fit
5. Candidate proposals	Outward cylinder/cone projection, multi-scale tubular score, surface NMS	Surface + CT -> high-recall ostium candidates	1 day	Adjacent bright structures	Require bidirectional radial boundary and path continuity
6. Local tracking	3-5 beam hypotheses, 0.5-1 mm steps, physical-space cost	Candidate -> proximal centreline <=10 mm	1 day	Gaps/artifacts	Fast marching within a candidate patch
7. Eligibility and topology	Path >=5 mm, first-bifurcation stop, cap rejection, ostium/path overlap graph	Candidate paths -> unique direct daughters	1 day	Overmerge nearby openings	Keep separate if wall-opening components are disconnected
8. Geometry	Signed-distance zero crossing, arc-length interpolation, tangent fit, orthogonal cross-section	Path -> ostium, seed, direction, radius	1 day	Partial volume at 1.5 mm	Blend equivalent-area and inscribed-radius estimates
9. Output validation	JSON schema + invariant assertions	Daughters -> strict prediction JSON	0.5 day	Evaluator mismatch	Minimal organiser fields only
10. Visualization	Plotly offline HTML + PNG panels; Slicer for development	CT/mask/result -> Aorta Map and evidence cards	1 day parallel	UI consumes core time	Static three-view PNG generator

6.2 Algorithm details that matter

Crop before resampling. Create a physical 20-25 mm margin around the aorta, then optionally resample only that ROI to 1.0-1.25 mm isotropic for stable geometry. Preserve the original image object and return outputs in its physical frame. On the 1.5 mm cohort, upsampling improves numerical consistency but cannot recover detail; confidence should reflect that.

Normalize from the parent. Erode the aorta by 1-2 mm to reduce partial volume. Estimate the core median, interquartile range, and MAD. Estimate background from shells at 2-5 mm and 8-15 mm, excluding extreme bone-like values. Construct an intensity likelihood rather than a hard threshold.

Propose from the wall. For each boundary location outside cap zones, follow the surface normal and a small cone of alternative directions. Sample central-cylinder intensity, radial contrast, and two or three vesselness scales over 0-7 mm. Max-pool over direction. Threshold permissively and use surface connected components/watershed to preserve nearby separate openings.

Track and validate. Advance in 0.5-1 mm physical steps. At each step score several directions in a forward cone and plausible radii. Keep a small beam with curvature and direction-change penalties. Accept only if the path reaches at least 5 mm with consistent central intensity and closed radial boundary evidence. Stop by 10 mm or at a stable two-way split.

Refine outputs coherently. Intersect a smoothed centreline with the mask signed-distance zero level to refine the ostium. Resample the centreline by arc length; the seed is exactly 5 mm. Fit the initial direction to points from roughly 1-5 mm to reduce single-step noise, orient it outward, and normalize. In the seed-normal plane, segment the connected lumen around the centre and report an equivalent circular radius $\sqrt{\text{area}/\pi}$, cross-checked against a maximum-inscribed radius. SlicerVMTK's cross-section tooling validates this style of measurement in development.^[20]

Deduplicate by evidence. Build a graph whose nodes are accepted candidates. Add an edge when two candidates share the same wall-opening component, lie within a small physical ostium distance, or their first 5 mm paths overlap strongly. Keep the best-supported node per equivalence group, except when two disconnected wall openings demonstrate separate origins. This directly implements the challenge's nearby-origin and common-trunk rules.

6.3 Confidence without expensive ensembles

Use a calibrated composite score from: proposal peak prominence, fraction of the first 5 mm with strong lumen likelihood, radial boundary completeness, radius stability, tracker best-vs-second-best margin, cap distance, and duplicate margin. On the development set, map the score to empirical precision. For the UI, show high/medium/low confidence and the evidence components. For scoring, choose one threshold based on the discovery F1 operating point.

6.4 Best variants

- Best accuracy-first approach: Pipeline A plus a tiny candidate-level learned reranker trained on reference ostia and hard negatives, with the deterministic tracker and geometry retained.
- Best speed-first fallback: Adaptive periaortic shell segmentation, cap removal, connected components, 5 mm persistence, line fit, and graph NMS.
- Best creative differentiator: The unwrapped Aorta Map linked to branch evidence cards and confidence.
- Best visualization stack: Offline Plotly HTML for judging; deterministic PNG checks for submission; 3D Slicer/SlicerVMTK only for development review.
- Best overall strategy: Ship a correct classical baseline on day one, then improve discovery with surface proposals and short tracking; add learning only if labelled evidence and time justify it.

7. Execution plan

7.1 Must-have core

1. Exact CLI and robust NIfTI loading for .nii and .nii.gz, including gzip-content sniffing.
2. Geometry validator: grid/transform match, physical-coordinate round-trip, deterministic output.
3. Adaptive shell baseline producing valid JSON on all 25 cases.
4. Cap exclusion, 5 mm persistence, common-trunk handling, and duplicate suppression.
5. Three automated visual checks with aorta mask, ostia, seeds, and direction arrows.
6. Runtime and peak-memory benchmark on the largest high-resolution and longest 1.5 mm cases.

7.2 High-value additions

1. Signed-distance surface normals and outward projection candidate map.
2. Multi-hypothesis 10 mm tracker.
3. Cross-sectional radius refinement at the seed.
4. Branch evidence cards and an interactive 3D overview.
5. Confidence score and failure-reason log.

7.3 Winning differentiators

1. Unwrapped Aorta Map with branch locations by arc length and angle.
2. Surface-opening graph that visibly explains nearby origins, common trunks, and deduplication.
3. Side-by-side robustness demo across contrast and resolution cohorts.
4. Optional learned candidate reranker with a clean ablation table.

7.4 Stretch features

1. Threshold-stability uncertainty halo.
2. Animated 0-10 mm branch traversal.
3. Proximal diameter profile and downloadable verification report.

4. Optional iliac extension behind a separate flag.

7.5 Parallel workstreams for a four-person team

Workstream	Owner focus	First deliverable	Integration contract
A. I/O and geometry	NIfTI, physical transforms, schema, tests	Valid JSON for a synthetic case	CaseGeometry, validator, deterministic serializer
B. Detection	normalization, surface proposals, tracker	Candidate paths and confidence	List of physical-space polylines with evidence
C. Evaluation	manual review protocol, proxy matcher, runtime/memory	Case dashboard and ablations	CSV/JSON metrics and failure labels
D. Product/demo	PNG checks, 3D view, Aorta Map, five-minute story	One clickable case	Reads only scored JSON + optional diagnostics

If the team is smaller, combine A+C and B+D. Do not let visualization modify detection outputs.

7.6 Suggested 36-hour build order

Hours 0-3: Lock the interpretation; ask organiser questions; implement loader, metadata audit, strict serializer, and synthetic geometry tests.

Hours 3-8: Build the adaptive shell baseline, cap suppression, path-length filter, and three-case visualizer. Run all 25 cases. This is the minimum survivable submission.

Hours 8-16: Add surface normals, outward projection scoring, and candidate clustering. Compare on manually reviewed cases representing high-resolution/high-contrast, high-resolution/low-contrast, 1.5 mm/high-contrast, and 1.5 mm/low-contrast conditions.

Hours 16-24: Add short beam tracking, ostium refinement, radius plane, and graph deduplication. Benchmark worst-case memory and runtime after every significant change.

Hours 24-30: Freeze the core. Build Aorta Map/evidence cards, README, one-command environment, and known-failure report. If reference labels exist, tune the final threshold on held-out cases.

Hours 30-36: Rehearse the five-minute demo, run from a clean offline environment, regenerate development predictions, and keep a frozen fallback tag.

7.7 Kill criteria

- Drop full-volume Frangi if it adds material time without a clear discovery gain.
- Drop any learned model that cannot beat the classical candidate precision/recall on held-out cases.
- Drop isotropic resampling if memory exceeds 2 GB or runtime loses the 60-second margin; use native physical sampling.
- Drop a full vessel mesh if segmentation leaks or VMTK requires manual seeds.
- Freeze visualization once it can select a branch and show its evidence; do not polish at the expense of core tests.

8. Risks and fallbacks

Risk	Early test	Mitigation	Fallback
Fixed thresholds fail across contrast	Compare aortic-core statistics across all cases	Per-case robust likelihood	Two-mode high/low enhancement parameters
Coarse 1.5 mm scans erase small vessels	Review smallest visible origins in subjects 16-25	Physical interpolation, wider uncertainty, radius-aware acceptance	Conservative minimum-size setting after organiser clarification
Bone/calcification creates false positives	Hard-negative gallery near vertebrae	Radial tube boundary, direction persistence, background contrast	Tighten 5 mm continuity requirement
Veins or organs mimic branches	Trace connection to the aortic wall and outwards	Wall-opening component + direct-parent topology	Raise proposal prominence threshold
Cropped mask caps are detected	Synthetic cap test and real endpoint gallery	Centreline endpoint bands and normal/tangent test	Exclude first/last 4-6 mm of aortic arc length
Nearby ostia merge	Synthetic two-opening phantom	Surface watershed and disconnected-opening rule	Reduce surface NMS radius
One trunk splits before 10 mm	Synthetic Y phantom	Count wall opening once; stop at first bifurcation	Keep shared proximal path as one branch
Branch mask gap fractures track	Blur/noise perturbation test	Hysteresis, beam search, gap allowance of 1-2 steps	Fast marching within local patch
Radius unstable	Compare equivalent-area and inscribed radius	Robust plane fit and connected seed component	Median radius from planes at 4.5-5.5 mm
Physical coordinates wrong	Round-trip known indices and transformed synthetic volume	SimpleITK transforms only	Fail fast rather than emit wrong JSON
Gzip content with .nii suffix fails reader	Regression tests on subjects 1 and 16	Magic-byte sniff and suffix normalization	Stream-decompress to temporary .nii
Runtime/memory exceeds limit	Profile largest voxel and longest-coverage cases	Crop first, float32/uint8, blockwise features	Pipeline C without vesselness
No references for tuning	Manual blinded review and synthetic phantoms	Keep parameters physical and few	Conservative precision-balanced threshold
UI consumes evaluation time	Separate --visualize command	Precompute after JSON	Static PNG only

8.1 Validation suite

Create synthetic cylinders attached to a parent tube with known physical transforms. Include: one branch, two close wall openings, one common trunk that bifurcates at 7 mm, a downstream-only branch, a cap-like continuation, an oblique branch, anisotropic spacing, rotated direction matrices, and gzip/uncompressed storage. Assert branch count, ostium error, seed arc length, direction angle, radius error, and duplicate count.

For real data, create a blinded review sheet rather than tuning by memory. Review at least eight representative cases with two teammates. Record only visible direct origins, confidence, cap/merge errors, and whether the first 5 mm is traceable. If official references arrive, use one-to-one Hungarian matching in physical space and mirror the organiser metric as closely as disclosed.

8.2 Reproducibility checklist

- One setup command and one run command exactly as required.
- Pinned versions and vendored/offline-available wheels or environment.
- No model or resource download at inference.
- Fixed random seed; deterministic candidate ordering.
- Peak-memory and wall-time log.
- Minimal JSON schema validation.
- Unit tests for continuous-to-physical and index-to-physical transformations.
- All thresholds expressed in millimetres, HU-relative statistics, or dimensionless scores.
- A clean run on a 4-core/8 GB constrained environment before submission.

9. Final winning strategy

If the only objective is to maximize the probability of winning, build Branchseed Navigator:

1. A CPU-first detector that uses the supplied aorta as an anatomical search anchor.
2. A self-calibrated surface projection that proposes every plausible wall opening without assuming names or counts.
3. A short multi-hypothesis tracker that must prove at least 5 mm of daughter lumen and stops by 10 mm or the first split.
4. A graph-based instance layer that explicitly handles duplicates, nearby origins, common trunks, caps, and downstream branches.
5. One coherent geometric estimator for ostium, 5 mm seed, direction, and radius in physical space.
6. A strict, boring JSON file for the evaluator.
7. A memorable Aorta Map and branch evidence cards for judges.

The five-minute demo should open with the constraint and the correction: "This is not a full segmentation problem and not MRI. It is a CPU-constrained instance-discovery problem on CTA." Then show one low-contrast 1.5 mm case, click an ostium on the Aorta Map, and reveal the linked CT evidence, 5 mm seed, direction arrow, and radius plane. Show the duplicate/common-trunk logic on a second case, the runtime/memory panel on a third, and close with a failure case plus the fallback. That story demonstrates technical depth, clinical usefulness, and operational honesty in the exact order the score rewards.

Do not compete on the size of the model or the gloss of the mesh. Compete on task alignment, topology, measurement correctness, and explainability.

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Appendix A. Starting parameters for a first implementation

These are initialization values, not claimed optimal settings. Express them as configuration and tune only against labelled or blinded-review evidence.

Parameter	Starting value	Rationale
Periaortic crop margin	25 mm	Covers the 10 mm scored path plus context and tracking error
Aortic-core erosion	1.5 mm	Reduces boundary partial volume while retaining samples
Candidate ray/cone length	7 mm	Must demonstrate most of the 5 mm eligibility path
Surface candidate spacing	1-1.5 mm	Close to the coarsest native resolution
Direction cone half-angle	45 degrees	Allows oblique departures while remaining outward
Tracker step	0.5-0.75 mm at 1 mm ROI spacing	Stable arc-length interpolation
Tracker beam width	3-5	Handles local ambiguity without global search cost
Maximum path	10 mm	Exact challenge limit
Acceptance path	≥ 5 mm	Exact eligibility rule
Initial cap exclusion band	5 mm along aortic arc	Avoids cropped endpoints; tune after organiser clarification

Parameter	Starting value	Rationale
Candidate merge radius	2-3 mm plus opening connectivity	Conservative deduplication; preserve separate wall components
Vesselness scales	0.8, 1.6, 2.8 mm	Small physical set spanning plausible proximal daughter radii
Cross-section sampling	0.25-0.5 mm in-plane	Improves radius fit without resampling the full volume

Appendix B. Output invariants

Before writing a case result, assert:

1. `case_id` is stable and nonempty.
2. Every `instance_id` is unique and matches `branch_###`.
3. Every `parent_instance_id` equals `aorta`.
4. Every coordinate contains three finite physical-millimetre values.
5. The seed lies 5.0 mm along the accepted path within a documented interpolation tolerance.
6. `radius_mm` is finite, positive, and within a broad sanity range.
7. `direction_xyz` contains three finite values, has norm within $1e-6$ of one after serialization, and points away from the parent.
8. No accepted ostium lies on an excluded cap component.
9. No pair of accepted daughters is classified as the same wall opening/path equivalence group.
10. Re-running the case produces byte-identical JSON except for deliberately excluded timing diagnostics.